

What each paper is for

The participant sheets are deliberately neutral: they paraphrase what each paper says about itself and ask questions, but do not say which papers are weak. The groups find that out. This key is for circulating facilitators, so that a group that is stuck can be nudged toward the point without being handed it.

How to nudge: ask the prompt again, then point at the relevant row of the appraisal checklist. Only if they are still stuck after the session on that principle, name the issue.

Group	Paper	Its job in the pack
1	Thiagarajan et al. 2026, sentinel node biopsy in early oral cancer	The well-built model , used to show that a sound analysis can still use a retired threshold rule, and that the abstract's own arithmetic needs checking.
2	Guleria et al. 2025, laparoscopic IPOM for ventral hernia	The paper that concludes not cost-effective , which is rare in print, and a clean example of a natural-unit ICER that cannot be judged against a per-QALY threshold.
3	Uy et al. 2021, surgery for rheumatic mitral valve disease	The clinically driven conclusion: the answer turns on which operation is chosen, and repair dominates current practice, yet current practice has not moved.
4	Kaur et al. 2022, population screening for diabetes and hypertension	The best paper in the pack for showing where an ICER really comes from: the conclusion flips on a health-system assumption, not a clinical one.
5	Sharma et al. 2016, Kangaroo Ward Care for very low birth weight infants	The deliberately weak paper. Titled a cost-effectiveness analysis; actually a cost comparison inside a trial. A good group takes it apart in fifteen minutes.
6	Patel et al. 2015, ward-based NIV for COPD respiratory failure	The technically competent paper. Right structure, lifetime, discounting, three kinds of sensitivity analysis; its weakness is where the numbers come from.
7	Gupta et al. 2022, peritoneal dialysis first	The analysis that ends in a negotiating position : a price at which the policy pays, rather than a yes or no.
8	Srinivasan et al. 2024, childhood AML at a tertiary centre	The real-world, DALY-based paper: honest data from one centre, a ratio that cannot be compared with the QALY papers, and a published peer appraisal to check the group against.

REPORT-BACK PAIRS

Pair	Groups	What the contrast should surface
1	1 and 2	a well-built model with a favourable verdict, against one that concludes 'not cost-effective'
2	4 and 3	a conclusion that turns on a health-system assumption, against one that turns on clinical choices
3	5 and 6	a paper that is not really a cost-effectiveness analysis, against a technically competent one
4	7 and 8	an analysis that ends in a price to negotiate, against one that ends in a cost per DALY

Four pairs at six minutes is 24 minutes. Keep one minute in hand. Cut a group off at two minutes; the contrast is the point, not the summary.

Thiagarajan et al. 2026, sentinel node biopsy in early oral cancer

Oncology and
surgery
pairs with Group 2

Thiagarajan S, Sharda S, Chugh Y, Gupta N, Pramesh CS, Prinja S. Sentinel lymph-node biopsy guided neck dissection versus elective neck dissection in the management of early-stage oral cancer: a cost-utility analysis. *Cancer Med* 2026;15(2):e71571. PMID 41644818 · PMC12875840 · doi:10.1002/cam4.71571

THE PAPER AGAINST THE CHECKLIST

Decision problem	Early oral SCC; three neck-management strategies. Clear.
Comparator	END alone and END with frozen section. Plausibly current practice in Indian cancer centres; ask the group whether END is what their own hospital does.
Perspective	Payer. The Reference Case asks for abridged societal in the base case, with payer reported separately.
Outcomes	QALYs. Check which utility source (Indian value set or borrowed).
Horizon, discounting	Lifetime; 3%. Meets the Reference Case.
Threshold	One times GDP per capita.
Uncertainty	One-way and PSA; 94% probability cost-effective.

WHAT A GOOD GROUP FINDS

- **The threshold rule is wrong in principle but harmless here.** The ratios are a few thousand to under twenty thousand rupees per QALY, far below GDP per capita and below India's willingness-to-pay estimate of ₹2,12,307 from Block 2. A good group says both things: the rule is retired, and it does not change this verdict. That is the mature position; the lazy one is to discard the paper for using it.
- **The abstract's arithmetic does not reproduce.** $5,564 \div 0.31$ is about 17,950 and $2,507 \div 0.33$ is about 7,600, not 8,088 and 16,709. The ratios look swapped, or computed from differently discounted figures. Send the group to the results table. Whatever the explanation, the lesson is: always recompute an ICER from the components.
- Payer perspective only. Out-of-pocket costs for travel to a centre with nuclear medicine may differ between strategies.

Before the day: Open the full text and confirm (a) which utility weights were used and (b) the explanation for the ICER arithmetic, so the facilitator can answer when the group asks.

Guleria et al. 2025, laparoscopic IPOM for ventral hernia

Oncology and
surgery
pairs with Group 1

Guleria C, Kumar D, Sahoo KC. Review for cost-effectiveness analysis of laparoscopic Intra-peritoneal Onlay Mesh for ventral hernia repair in Indian settings. *Cost Eff Resour Alloc* 2025;23(1):27.

PMID 40495189 · doi:10.1186/s12962-025-00638-4

THE PAPER AGAINST THE CHECKLIST

Decision problem	Laparoscopic IPOM vs open repair for ventral hernia, Indian health system.
Comparator	Open repair. Probably still current practice in most public hospitals.
Effectiveness	Meta-analysis of 10 RCTs. The strongest evidence source in the pack.
Costs	NHSCD. Health system only; no out-of-pocket.
Outcomes	Natural units: recurrence (primary), wound infection. No QALYs.
Horizon, discounting	Five years. We could not find a discount rate on our read; ask the group to look.
Threshold	INR 2,14,000, per-capita income.
Uncertainty	PSA, 10,000 simulations; 54.0% cost-effective.

WHAT A GOOD GROUP FINDS

- **Unit mismatch.** A per-capita-income threshold is a (retired) rule of thumb for cost per DALY or QALY. Comparing *rupees per wound infection averted* against it has no meaning. A natural-unit ICER cannot enter a league table or meet a threshold.
- **Outcome choice.** Recurrence was primary and showed no difference; the ICER is built on a secondary outcome that did. Not necessarily wrong, but it must be justified, and the group should notice it.
- **54% is a coin toss.** 'Less likely to be cost-effective' is not what 54% says. The honest reading: the analysis cannot tell. Which fits Block 3C's point that a probability of being cost-effective is not the probability of being right.
- A rare published negative. Ask the room how many 'not cost-effective' Indian evaluations they have read. Publication bias exists here too.

Before the day: The handover describes this as an HTAIn study. On our read the paper mentions HTAIn and one author is at the Department of Health Research, but we did not confirm it was commissioned through HTAIn. Say 'HTAIn-linked' only if the full text confirms it.

Uy et al. 2021, surgery for rheumatic mitral valve disease

Cardiology and
medicine
pairs with Group 4

Uy J, Ketkar AG, Portnoy A, Kim JJ. Cost-utility analysis of heart surgeries for young adults with severe rheumatic mitral valve disease in India. *Int J Cardiol* 2021;338:50-57. PMID 34090957 · doi:10.1016/j.ijcard.2021.05.059

THE PAPER AGAINST THE CHECKLIST

Decision problem	Young adults, severe RMVD, choice of operation. Clear and clinically framed.
Comparator	A modelled practice mix (32/33/35). Defensible, but its source should be checked.
Perspective	Indian public payer only.
Outcomes	QALYs, lifetime.
Costs	Reported in US dollars.
Threshold	India's GDP per capita, \$2,005.
Uncertainty	Not described in the abstract. The planning notes list one-way and probabilistic analysis; confirm.

WHAT A GOOD GROUP FINDS

- **Dominance is not the whole answer.** Repair dominates standard of care and mechanical replacement. But bioprosthetic replacement gains 0.4 QALY more than repair at \$1,590 per QALY. So 'repair is optimal' holds only if the threshold is below \$1,590. At the GDP rule the authors cite, bioprosthetic would be preferred. The conclusion depends on a threshold the abstract leaves ambiguous. A good group spots this.
- Check the arithmetic: $\$660 \div 0.4 \text{ QALY}$ is \$1,650, close to the reported \$1,590 once QALYs are unrounded.
- **The practice gap.** If repair dominates, why do centres not do it? Surgical skill, learning curve, durability in rheumatic valves, follow-up access. These are real constraints the model may not carry; the ICER does not tell a surgeon how to acquire the skill.
- Mechanical valves at 6.2 QALYs: ask what drives that. Anticoagulation monitoring in young Indian adults, probably. A structural assumption worth naming.

Before the day: This paper is not open access and the abstract does not describe the sensitivity analysis. If anyone on the faculty has access, confirm what uncertainty analysis was done and the cost year before the report-back.

Kaur et al. 2022, population screening for diabetes and hypertension

Cardiology and
medicine
pairs with Group 3

Kaur G, Chauhan AS, Prinja S, Teerawattananon Y, Muniyandi M, Rastogi A, Jyani G, Nagarajan K, Lakshmi P, Gupta A, Selvam JM, Bhansali A, Jain S. Cost-effectiveness of population-based screening for diabetes and hypertension in India: an economic modelling study. *Lancet Public Health* 2022;7(1):e65-e73. PMID 34774219 · doi:10.1016/S2468-2667(21)00199-7

THE PAPER AGAINST THE CHECKLIST

Decision problem	Population screening for diabetes and hypertension: ages, frequencies, tests. Carefully specified.
Comparator	Current opportunistic detection. Exactly what the Reference Case asks for.
Perspective	Societal, including out-of-pocket.
Costs	Primary Indian data plus NHSCD, CHSI, national survey, package rates.
Outcomes	QALYs, EQ-5D-5L in 962 Indian patients.
Horizon, discounting	Lifetime from 30; 3%.
Threshold	One times GDP per capita.
Uncertainty	PSA, and a scenario analysis on treatment coverage that decides the result.

WHAT A GOOD GROUP FINDS

- **The flip.** Screening that finds people who are then not treated produces cost and no health. The ICER is a statement about the health system downstream of the test, not the test. This is the single most transferable lesson in the pack.
- Against the Uy paper: there the answer turns on the surgeon's choice; here on whether a Health and Wellness Centre dispenses the drugs.
- It uses the retired GDP rule too. Here, unlike Thiagarajan, it matters: the ICERs sit near the threshold, so the rule is doing work.
- Near-model Reference Case compliance: comparator, perspective, Indian utilities, primary costing, PSA. Use it as the 'what good looks like' exhibit.

Before the day: Confirm open access at the journal site before telling the group they can download it.

Sharma et al. 2016, Kangaroo Ward Care for very low birth weight infants

Critical care, anaesthesia,
neonatology
pairs with Group 6

Sharma D, Murki S, Oleti TP. To compare cost effectiveness of 'Kangaroo Ward Care' with 'Intermediate intensive care' in stable very low birth weight infants (birth weight < 1100 grams): a randomized control trial. *Ital J Pediatr* 2016;42:64. PMID 27412638 · doi:10.1186/s13052-016-0274-3

THE PAPER AGAINST THE CHECKLIST

Decision problem	Early kangaroo ward vs intermediate care, stable VLBW infants. Clear.
Comparator	Intermediate intensive care. Reasonable current practice.
Costs	Hospital top down, parents bottom up. Promising, and a useful link to the costing session.
Outcomes	Cost and length of stay. No QALY, no DALY, no ICER.
Horizon	Ends at discharge.
Discounting	Not reported; not needed at this horizon.
Uncertainty	No sensitivity analysis reported.

WHAT A GOOD GROUP FINDS

- **Not a cost-effectiveness analysis.** With no health outcome in the ratio it is a cost analysis (Block 1, box 4). If the claim is 'same outcome, lower cost', it is a **cost-minimisation** argument, which is legitimate only if equivalence of outcomes is *demonstrated*. Ask what the paper shows about neonatal outcomes, and whether the trial was designed to show equivalence.
- The title is the teaching point: a word in a title is not a method. Link to Block 1's 'three errors we will keep catching'.
- Credit where due: a randomised design, and patient-side costs collected bottom up, which many better-known papers omit.
- Horizon at discharge misses readmission, growth and neurodevelopment; the outcomes that would matter over a lifetime are not in scope.

Patel et al. 2015, ward-based NIV for COPD respiratory failure

Critical care, anaesthesia,
neonatology
pairs with Group 5

Patel SP, Pena ME, Babcock CI. Cost-effectiveness of noninvasive ventilation for chronic obstructive pulmonary disease-related respiratory failure in Indian hospitals without ICU facilities. *Lung India* 2015;32(6):549-556. PMID 26664158 · PMC4663855 · doi:10.4103/0970-2113.168137

THE PAPER AGAINST THE CHECKLIST

Decision problem	Ward NIV vs standard care, COPD respiratory failure, hospitals without ICU. Well defined.
Comparator	Standard treatment without NIV. Genuinely current practice in such hospitals.
Perspective	Societal.
Effectiveness	Published literature, not Indian data. Much likely from ICU or high-income settings.
Outcomes	QALYs; survival.
Horizon, discounting	Lifetime; 3% stated for future costs. Check whether QALYs were discounted.
Threshold	GDP per capita, \$1,489 (2012).
Uncertainty	One-way, two-way, probabilistic. 100% preferred above \$250 per QALY.

WHAT A GOOD GROUP FINDS

- **Average versus incremental**, live. The \$78 and \$75 figures are average cost-effectiveness ratios, total cost over total QALYs, which Block 3B calls meaningless for a decision. The ICER is $\$101 \div 1.67 =$ about \$61. Ask the group which number a decision-maker should read.
- **Transferability**. Every input is borrowed. NIV effectiveness on a general ward in an Indian district hospital, with its staffing, may be far from the trial settings. PSA varies parameters inside their borrowed ranges; it cannot test whether the ranges apply here. That is structural and transferability uncertainty, which Block 3C says PSA does not handle.
- **When the answer is obvious, the question moves**. At \$61 per QALY no threshold matters. The real barrier is feasibility: staff, masks, monitoring on a ward. The economics says 'yes'; implementation is the open question.
- 2012 US dollars against 2012 GDP. Ask how they would update it for a decision today.

Gupta et al. 2022, peritoneal dialysis first

Gupta D, Jyani G, Ramachandran R, Bahuguna P, Ameel M, Dahiya BB, Kohli HS, Prinja S, Jha V. Peritoneal dialysis-first initiative in India: a cost-effectiveness analysis. *Clin Kidney J* 2022;15(1):128-135.
PMID 35035943 · PMC8757426 · doi:10.1093/ckj/sfab126

THE PAPER AGAINST THE CHECKLIST

Decision problem	PD first vs HD first for kidney failure; real-world and PMNDP scenarios. Policy-framed.
Comparator	HD first. Current practice in India.
Perspective	Societal, including indirect (productivity) costs in the base case.
Costs	PMNDP data and micro-costing; out-of-pocket from patient interviews; indirect costs by the human capital approach.
Outcomes	QALYs and life years. EQ-5D in 192 patients, valued with the Thai tariff because no Indian value set existed then.
Horizon, discounting	Lifetime; 3%. One-year cycles.
Threshold	Per-capita GDP, INR 1,48,171 (2019).
Uncertainty	PSA, 1,000 iterations, gamma for costs and beta for probabilities; threshold analysis on consumable price.

WHAT A GOOD GROUP FINDS

- **The perspective decides the verdict.** Cost-saving from the societal perspective, which includes productivity costs. On direct costs only, cost-effective only at INR 70 per unit. The Indian Reference Case (2023, after this paper) puts indirect costs in *sensitivity analysis only*. A good group sees that the headline would be weaker under today's rules.
- **A price, not a verdict.** The threshold analysis tells a purchaser what to negotiate for. That is often the most useful output an economic evaluation can give, and it links to the costing session: prices are not costs, and prices can move.
- **Borrowed utilities.** Thai tariff, because India had no EQ-5D value set at the time. Now it does (Block 3B). Legitimate then; a flaw if repeated now.
- Against Srinivasan: both are Indian, both real-world, but this one ends in a number a ministry can act on, the other in a cost per DALY that cannot be compared with anything else in the pack.

Srinivasan et al. 2024, childhood AML at a tertiary centre

Paediatrics, infectious disease,
nephrology
pairs with Group 7

Srinivasan S, Bolous NS, Batra A, Bharti S, Singh N, Shaikh T, Yadav A, Kanwar V. Cost-effectiveness of treating childhood acute myeloid leukemia at a tertiary care center in North India. *Pediatr Blood Cancer* 2024;71(11):e31242. PMID 39126354 · doi:10.1002/pbc.31242

THE PAPER AGAINST THE CHECKLIST

Decision problem	Treating childhood AML with curative intent at one centre. The alternative is implicit.
Comparator	Not stated in the abstract; in effect no curative treatment. Ask the group whether that is current practice.
Perspective	Hospital costs from the database. Out-of-pocket and family costs unclear.
Effectiveness	Observed outcomes in 59 children, one centre, retrospective.
Outcomes	DALYs averted. The Reference Case permits DALYs only under age 4; these are children up to 15.
Horizon	Three-year survival observed; DALYs imply extrapolation to a lifetime. Check how.
Threshold	Cost per DALY averted as a share of GDP per capita (24%).
Uncertainty	None described in the abstract.

WHAT A GOOD GROUP FINDS

- **Not comparable.** A cost per DALY averted cannot sit in a league table beside cost per QALY gained. Paired with Gupta, which ends in a price a ministry can act on, this one ends in a ratio that answers 'is treatment worth offering at all?', which is a real and important question, but a different one.
- **An implicit comparator.** 'Cost-effective' compared with what? Almost certainly no curative treatment, where DALYs averted are measured against near-certain death. That makes the ratio favourable almost by construction.
- **Denominators.** With abandonment and 30% treatment-related mortality, whether the cost per patient includes the children who died early or left treatment changes the answer. Ask the group what the paper divides by.
- Credit: real Indian costs and outcomes, and the authors judge against a stringent threshold rather than the GDP rule.
- **The Comment.** The group can compare its appraisal with a published one. Treat it as a model of peer critique, not an answer key.

Before the day: (a) Read the published Comment (Sra, Bakhshi, Ganguly, *Pediatr Blood Cancer* 2025;72(1):e31305, PMID 39228042) before the day, so you can say what it argued. (b) The paper is open access (CC BY-NC-ND): print the full text for Group 8.